

Donald Khek

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EDUCATION

University of Florida

Bachelor of Science in Electrical Engineering

Gainesville, FL

Expected May 2028

- **Cumulative GPA:** 3.89/4.0
- **Coursework:** C++ for Engineers, Computer Aided Graphics and Design, Digital Logic and Computer Systems, Signals and Systems

EXPERIENCE

TEA Lab

Jun. 2026 – Present

Undergraduate Research Assistant | Python, NumPy, HiPerGator HPC

Gainesville, FL

- Engineered parallelized data generation on the **HiPerGator HPC Cluster** using Python and highway-env, which was used to train world models and baselines
- Proposed replacing unstable visual heuristics (bounding boxes/color markers) with simulator-native Time-to-Collision (**TTC**), establishing a robust ground-truth collision metric for the team
- Designed and trained a **PyTorch** ResNet-18 model to infer continuous TTC directly from images, achieving a high-precision **0.0035 MSE**

PROJECTS

Custom E-Paper E-Reader | C++, KiCad, ESP32, Hardware Design, SPI, I2C

- Engineered a 4-layer PCB in **KiCad** to route a dense component layout in a pocket form-factor, isolating noisy high-voltage E-Ink boost switching from **ESP32-C3** RF and SPI signal lines
- Implemented an automated **P-MOSFET** power-path circuit and **MCP73831** charge controller, achieving a **65–75 μ A** deep-sleep quiescent current to ensure multi-week battery standby
- Overcame a **4MB flash limitation** by refactoring **CrossPoint (C++)** firmware designed for 16MB chips, repartitioning memory and offloading fonts/assets to microSD over SPI

Personal Home-Lab & Network Infrastructure | Debian, Raspberry Pi, Docker

- Architected a dual-node home-lab using a **Raspberry Pi** and **Wake-on-LAN** to sleep-gate a **Debian** host, cutting baseline idle power by **>85% (from ~50W to <3W)**
- Migrated bare-metal Windows workloads to headless **Debian** via **Docker** and **Pterodactyl**, reclaiming **~3GB** of idle RAM while enforcing per-container cgroup limits
- Automated edge ingress using **Cloudflare DDNS** with split CNAME records, resolving local SSL/TLS loopback issues and enforcing **UFW** rules to isolate internal web panels from public game traffic

Gatorlator (Chrome Extension) | GitHub | Javascript, Chrome API, Web Audio

- Served as technical lead for a 4-person team, architecting a real-time tab-audio translation pipeline that won **Best Education, Accessibility, & Social Impact Project** at SwampHacks XII
- Achieved **<500ms translation latency** by streaming raw tab audio over **WebSockets** to **Deepgram**, pipelining intermediate transcripts for zero-lag subtitles alongside sentence-chunked **ElevenLabs TTS**
- Engineered continuous Web Audio capture within **Chrome Manifest V3**, deploying an offscreen document to bypass service-worker lifecycle termination and manage audio stream routing

TECHNICAL SKILLS

Languages : Python, JavaScript, HTML/CSS, Lua, C++, LaTeX

Hardware & Tools : KiCad, SolidWorks (CSWA Certified), Git, Docker, SPI, I2C, HiPerGator HPC (SLURM)

Frameworks & Libraries : PyTorch, NumPy, Web Audio API, Chrome Extensions (MV3), REST APIs

Operating Systems : Windows, Linux, Unix/macOS